

Observability Retention Capacity

July 2026 retention review | Owner: Leah Ortiz, Observability SRE | Next review: 2026-08-24

Decision at a glance

Decision

Keep 21-day hot logs, reduce hot trace retention to ten days, and finance cold-tier growth. The plan protects incident evidence while making sampling a deliberate capacity choice rather than an unbounded default.

Planning peak

19.6 TB/day

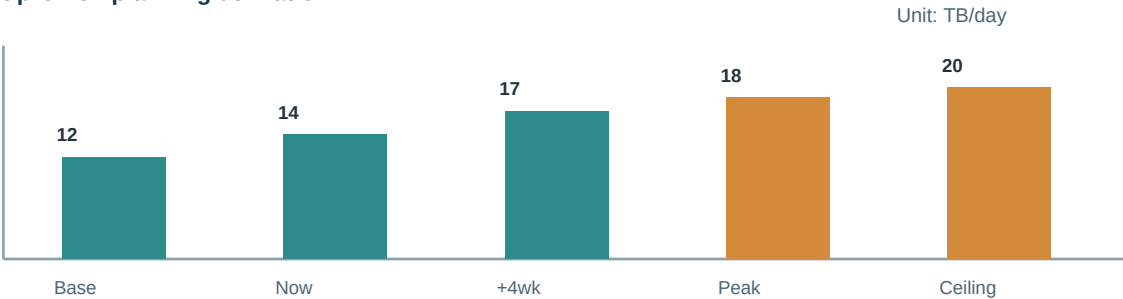
Evidence

High; ingest metering reconciled with cold-tier manifests

Scope and assumptions

This review separates logs, metrics, and traces because their retention cost and diagnostic value differ. The model assumes 21-day hot logs, 13-month metrics, 10-day hot traces, and a 17% trace-volume rise from identity rollout work. It excludes debug sampling requested during one-off investigations.

Forecast profile - planning utilization



Capacity signals

Measure	Current	Planning limit	Decision signal
Daily ingest	14.2 TB/day	19.6 TB/day	Move trace policy when ingest exceeds 17 TB/day.
Hot-tier fill	68%	< 78%	Keep enough room for a seven-day incident burst.
Trace sample rate	8%	<= 10%	Higher sampling needs storage approval, not an alert tweak.
Cold-tier transfer	2.4 TB/day	>= 2.0 TB/day	Slow transfer shortens hot retention first.

Plan, controls, and ownership

Action plan

Action	Due	Owner	Acceptance evidence
Trace policy	Aug 02	Observability team	Set traces to 10-day hot retention.
Cold allocation	Aug 11	Storage partner	Fund cold-tier capacity through October.
Sampling review	Aug 18	Service owners	Remove high-cardinality labels from checkout traces.
Retention audit	Aug 24	Leah Ortiz	Confirm logs retain 21 days after policy change.

Risk register

Risk	Likelihood	Impact	Response
High-cardinality labels	Medium	High	Block new labels without cardinality estimate.
Cold transfer backlog	Medium	Medium	Prioritize logs over traces during backlog recovery.
Incident sampling surge	Low	High	Time-box emergency sampling to 24 hours.

Decision rationale

Keep 21-day hot logs, reduce hot trace retention to ten days, and finance cold-tier growth. The plan protects incident evidence while making sampling a deliberate capacity choice rather than an unbounded default.

Selective cross-references

Review alongside: Identity capacity plan and Postgres storage growth plan. These linked plans are relevant because they share a limiting dependency or coordinated operating window.

Model notes: all values are synthetic planning figures. No customer records, production identifiers, secrets, or routable network addresses are included. Northstar Transit Cloud | Telemetry volumes are synthetic